

Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

Lewei Xu

Supervised by Dr. Yihao Ding · Dr. Daochang Liu



Table of Contents

- Introduction
- Survey of MP-VRDU
- Empirical Attribution Study
- Conclusion
- Proof of Concept: A Simple Method

Multi-Page Visually Rich Documents

Documents carry meaning through **text**, **visual information** and **layout** together. **Multi-page** documents add another layer: meaning can run across pages, as when a sentence refers to a figure several pages away.

Figure 10: Consequences of Genomic Imprinting Defects

Figure 10: Consequences of Genomic Imprinting Defects. The diagram illustrates the process of genomic imprinting, where certain genes are expressed or silenced based on whether they are inherited from the mother or father. It shows how defects in this process can lead to various developmental and functional issues, such as growth retardation, developmental delay, and specific clinical syndromes like Prader-Willi and Angelman syndromes. A photograph of a child is included to show the clinical context of these conditions.

Research paper
born-digital

– **Text.** Typeset or handwritten, embedded in the file or only visible in a scan.

Define your problem

Page # 1

Redacted by DOI

A scanned image of a handwritten form. The form has several sections with checkboxes and lines for text. Some parts are redacted with black boxes. A signature is visible at the bottom right. The text is written in cursive.

Signed form
scanned, handwritten

Parts and Features

Below are illustrations of your appliance with the parts and features called out. Your model may have all of the features shown and it may not be exactly as illustrated.

Style 1

A detailed diagram of a refrigerator with various parts and features labeled. The labels include: Automatic ice maker with crushed ice, Light, EZ-Track, Model and serial number label, Utility compartment, Wire rack, Egg bin, Removable freezer shelves, Adjustable slide-out freezer baskets, Snack bin, Slide-out freezer baskets, Moist drawer, Adjustable door bins, Leveling rollers, Base grille, Control panel, Light, EZ-Track, Model and serial number label, Utility compartment, Wire rack, Egg bin, Adjustable slide-out freezer shelves, Adjustable slide-out freezer baskets, Moist drawer, Adjustable door bins, Leveling rollers, Base grille, Crisper light, Humidity-controlled crisper, Door bin slides, Deep door bins.

Technical manual
layout-dependent

– **Visual information.** Charts, figures, photographs and handwriting.

Table of Contents

Section Financial Data

The following selected consolidated financial data is not necessarily indicative of results of future operations and should be read in conjunction with Item 7, Management's Discussion and Analysis of Financial Condition and Results of Operations and Item 8, Financial Statements and Supplementary Data. The following amounts related to earnings per share and share outstanding have been adjusted for the Stock Split for all periods reported, see Note 8 of Item 8, Financial Statements and Supplementary Data for further detail on the Stock Split.

Consolidated Statements of Operations:

	2012	2011	2010	2009	2008
Revenues	\$ 6,779,311	\$ 5,566,618	\$ 4,374,342	\$ 3,489,282	\$ 3,294,377
Operating income	303,626	402,448	228,347	419,907	276,946
Net income	127,641	266,799	112,493	173,112	226,126
Earnings per share:					
Basic	\$ 0.29	\$ 0.61	\$ 0.28	\$ 0.54	\$ 0.61
Diluted	\$ 0.28	\$ 0.62	\$ 0.26	\$ 0.54	\$ 0.59
Weighted average common shares outstanding	437,889	428,144	407,381	318,648	369,919
Diluted	436,416	401,894	427,817	411,227	381,181

Consolidated Statements of Cash Flows:

	2012	2011	2010	2009	2008
Net cash (used in) provided by operating activities	\$ (769,439)	\$ 36,483	\$ 97,831	\$ 21,586	\$ 317,732
Free cash flow (1)	(920,517)	(226,689)	(26,340)	(18,515)	185,519

(1) See "Liquidity and Capital Resources" for a definition of "free cash flow" and a reconciliation of "free cash flow" to "net cash (used in) provided by operating activities."

Consolidated Balance Sheets:

	2012	2011	2010	2009	2008
Cash, cash equivalents and short-term investments	\$ 2,310,115	\$ 1,808,496	\$ 1,200,491	\$ 748,078	\$ 797,811
Total current assets, net (2)	1,718,511	4,519,400	3,555,564	2,514,009	2,522,619
Working capital (3)	1,802,216	3,263,889	2,811,689	1,551,887	1,951,780
Total assets (3)	\$2,362,271	\$2,042,289	\$1,464,021	\$1,041,781	\$1,062,111
Long-term debt (4)	\$2,713,362	\$81,889	\$65,462	\$95,782	\$95,223
Long-term debt due to related party (1)	—	—	—	\$18,209	\$17,792
Total current liabilities	\$2,636,140	\$1,575,082	\$1,341,560	\$1,076,422	\$789,632
Total current liabilities	\$2,636,140	\$1,575,082	\$1,341,560	\$1,076,422	\$789,632
Total liabilities	\$2,636,140	\$1,575,082	\$1,341,560	\$1,076,422	\$789,632
Total assets	\$2,362,271	\$2,042,289	\$1,464,021	\$1,041,781	\$1,062,111

(1) Certain prior year amounts have been reclassified to conform to the current year presentation in the consolidated financial statements. See Note 1 and Note 2 in Item 8, Financial Statements and Supplementary Data for further detail.

Financial report
dense tables

– **Layout.** How information is arranged in the two-dimensional space of a page.

Leave a lasting impression.

Camry SE Hybrid

Seating details

Floating touchscreen display

With Camry's powerful stance and available black front grille with sport mesh insert, announce your arrival with authority. At night, you'll immediately notice Camry SE's LED headlights and fully integrated LED Daytime Running Lights (DRLs) that add an illuminating touch.

With Camry's perfectly placed steering wheel controls, everything you want is right at your fingertips. The floating display with an available 8-in. multimedia touchscreen offers Apple CarPlay™ and Android Auto™ compatibility with applicable smartphone devices.

What is Document Understanding?

Answering natural-language questions: **identify the relevant content** and **connect the evidence**, within one page or across pages.

LAYOUT ANALYSIS

3
PEW RESEARCH CENTER

Latinos Increasingly Confident in Personal Finances. See Better Economic Times Ahead
Yet many economic indicators show few gains for the community since the Great Recession

Latinos have become considerably more upbeat about their personal finances and optimistic about their financial future since the Great Recession, according to newly released results from a national survey of Latino adults. The survey also shows that Latinos have pulled even with the general U.S. population in their views of their personal finances and continue to outpace them on optimism about the future. However, community economic indicators show limited progress since the Great Recession.

Four-in-ten Latinos say their personal finances are in "excellent" or "good" shape, a 17 percentage point increase since 2008, when only 23% made a similarly positive assessment of their finances (the Great Recession began in December 2007).¹ By contrast, the share of all Americans who have a similarly positive view of their finances remained essentially flat during the same seven-year period (41% in 2008 vs. 43% in 2015).

The survey also shows that Hispanics are more upbeat in their financial expectations for the upcoming year than they were in 2008. About eight-in-ten Hispanic adults (80%) say they expect their family's financial situation to improve in the next year, up from 67% who said the same in 2008. By comparison, the U.S. public is not as upbeat –

Hispanics more confident in personal finances than in 2008, see better economic times ahead

% who say their personal financial situation is in "excellent/good shape"

% who expect their family's financial situation to improve "a lot" or "some" in the next year

General public

Hispanic

2004 2008 2011 2015

RECESSION

Source: Shaded area marks the Great Recession, which ran from December 2007 to June 2009.

Source: National Survey of Latinos conducted Oct. 23-Nov. 30, 2015; surveys conducted Dec. 8-13, 2010; Oct. 24-28, 2012; Sept. 12-16, 2012; Dec. 7-11, 2011; Dec. 3-8, 2008; Oct. 19-22, 2006; Aug. 5-10, 2004.

¹Latinos Increasingly Confident in Personal Finances, See Better Economic Times Ahead

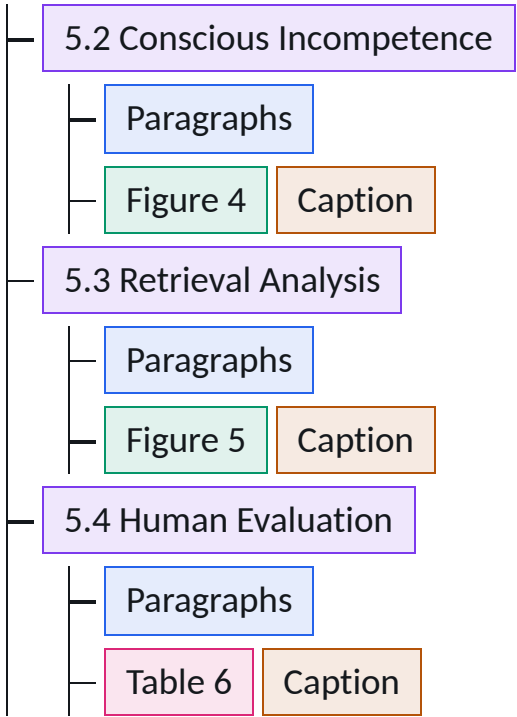
PEW RESEARCH CENTER

www.pewresearch.org

Title Section heading Text Figure / chart Table Caption / note Header / footer

DOCUMENT OUTLINE

Research paper, 24 pages



EXAMPLE QUESTION

“Does the model rated best by humans in Table 6 also have the best citation recall in Table 3?”

Evidence needed:

Table 6, this page

Table 3, previous page

Cross-page: it compares two tables on different pages, so neither alone is enough.

LLM Assistants

Provide a demonstration of how you would use tool calls to answer a difficult question over this long visually rich document.

Designing a sample tool-based walkthrough for a cross-page question.

I'll demonstrate on a question that needs a figure and a table: "**How often does the final pipeline stop at its earliest allowed point, and how does its page usage compare with the number of gold evidence pages?**" First I'll map printed page numbers to PDF pages, then read the relevant pages as images.

Read a file, updated tasks ▾

Added task Locate pages for Figure 4.2 and Table 4.2

Request

Map printed page numbers to PDF pages using the contents list and the offset seen on page 27

Added task Read figure and table pages visually

Request

Description	Read PDF pages 60-61 and extract the bar heights and token statistics
-------------	---

Added task **Verify and answer** ▾

Request

Question	Answer	Description
1. The number of people who are not in the category of 'Other' is 100. What is the number of people who are in the category of 'Other'?	100	Cross-check figure values against surrounding text, then answer

Completed task Locate pages for Figure 4.2 and Table 4.2

Started task Read figure and table pages visually

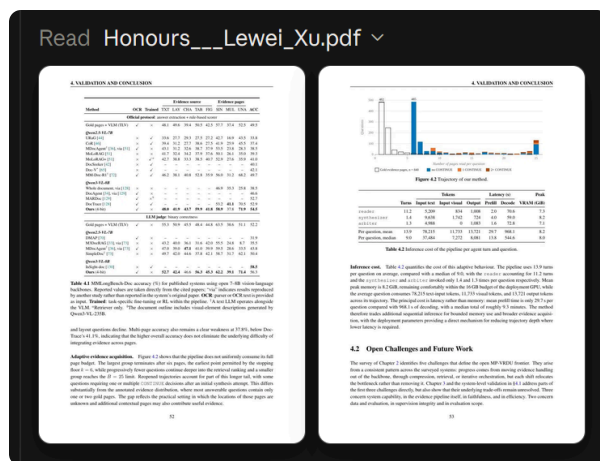
Read Honours Lewei Xu.pdf >

Verifying the data points plotted across the chart.

Completed task Read figure and table pages visually

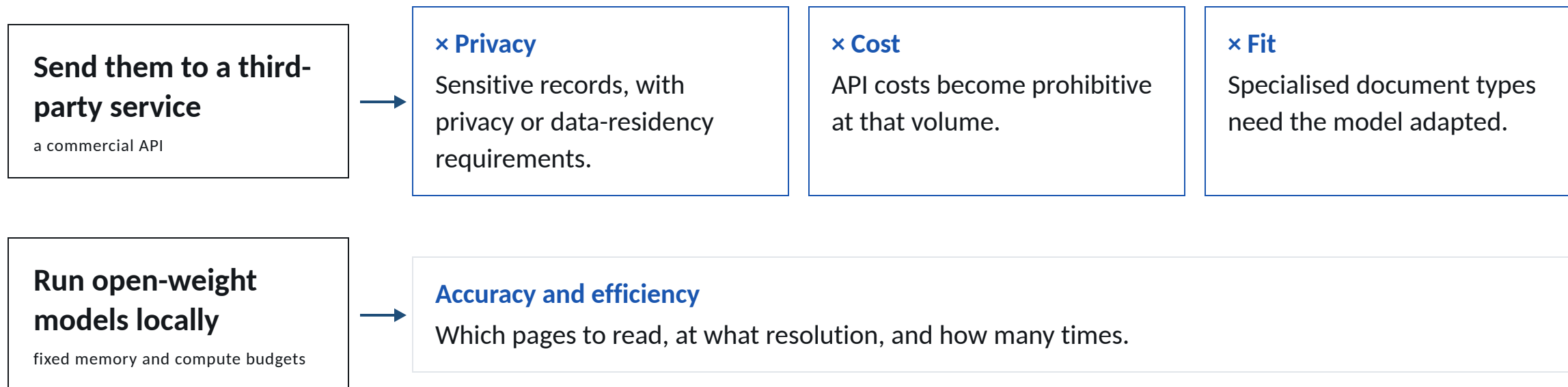
Completed task **Verify and answer**

- Upload a document to an assistant and **ask it questions**.
- My thesis, almost **100 pages**, and a difficult question: it still did well.
- An **agentic workflow**: it plans, locates the relevant pages, and inspects them before answering.
- It **renders pages as images**, reading visually as well as through text.
- Research reaching consumer products, but only one part of how document understanding is used in practice.



Real-World Constraints

In industry: classify, summarise or extract information from **thousands, or even millions, of documents**.



The central question: how can we improve document understanding **under a fixed compute budget**?

Table of Contents

- Introduction

- **Survey of MP-VRDU**

Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding.

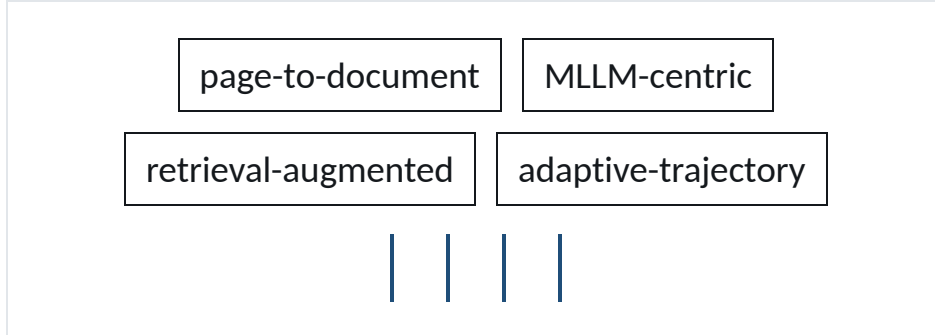
Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu

[EMNLP 2026, main conference](#)

- Empirical Attribution Study
- Conclusion
- Proof of Concept: A Simple Method

Survey: Research Gap and Contributions

RESEARCH GAP



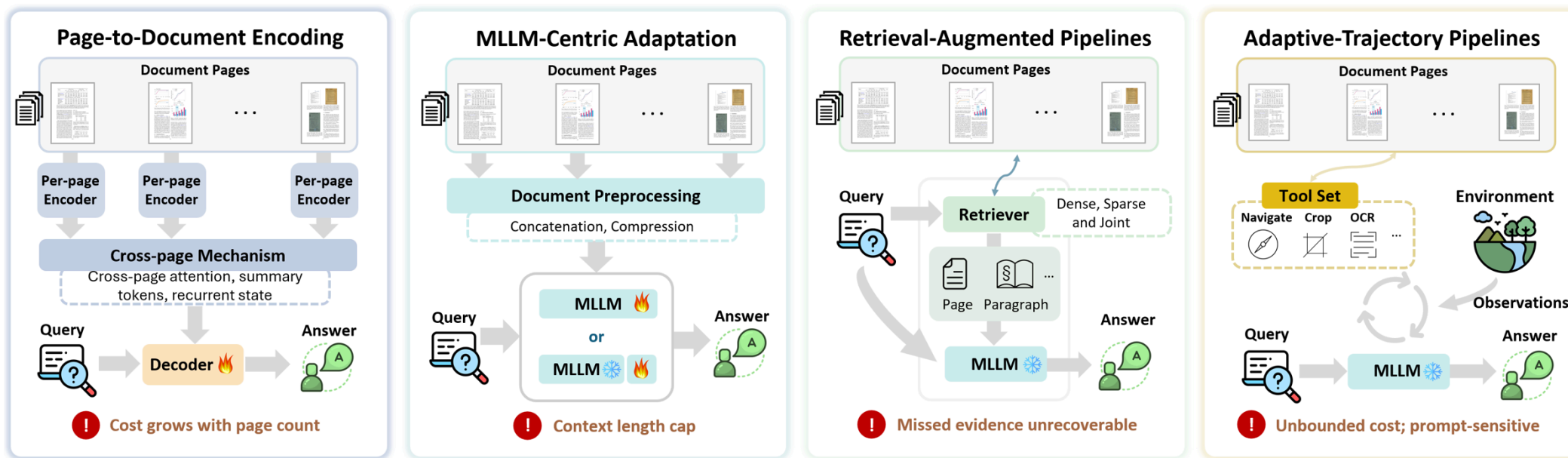
Multi-page systems developed along **parallel lines**, with different terminology and little comparison between them.

CONTRIBUTIONS

- 1 Brings these systems together through **evidence management**: how a system manages the relevant evidence across a very long document.
- 2 Organises them into **four architectural families**.

Multi-Page Architectural Overview

Four architectural families. They emerged roughly in this order, although their development overlaps.



Late 2023 to mid 2024

Encode each page, connect them with a cross-page mechanism, train end to end

Early 2024 to mid 2026

Feed all pages into a pretrained model's context, with further training

Early 2024 to mid 2026

Retrieve evidence first, then answer. The workflow is fixed in advance

Early 2025 to mid 2026

A language model decides what to inspect next from what it has found

Snowflake indicates typically frozen, fire indicates typically trained

Surveyed Systems at a Glance

Model	Venue	OCR	Vision Encoder	LLM Backbone	Mod.	Search	Reasoning	Agentic
Page-to-Document Encoding Models								
Arctic-TILT (2025)	ACL	Yes	U-Net	T5-Large	T, V, L	-	-	-
GRAM (2024)	CVPR	Yes	DocFormerv2	DocFormerv2	T, V, L	-	-	-
Hi-VTS (2023)	PR	Yes	DiT	T5-Base	T, V, L	-	-	-
RM-T5 (2024)	DAS	Yes	DiT	T5-Base	T, V, L	-	-	-
MLLM-Centric Adaptation Models								
DocR1 (2026)	AAAI	No	Qwen2.5-VL	Qwen2.5-VL-7B	V	-	CoT, HD	-
DocSeeker (2025)	CVPR	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B-Instruct	V	-	CoT, HD	-
Leopard (2024)	TMLR	No	SigLIP-SO400M	LLaMa-3.1-8B, Mistral-7B	V	-	CoT	-
mPLUG-DocOwl2 (2025)	ACL	No	ViT-L/14	mPLUG-Owl2	V	-	-	-
URaG (2026)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B / Qwen2.5-VL-7B	V	-	HD	-
Docipilot (2025b)	CVPR	No*	InternViT-300M	Multiple (Intern)	V	-	-	-
Texthawk2 (2024)	preprint	No*	SigLIP-SO400M	Qwen2.5-VL-7B-Instruct	T, V, L	-	-	-
DocSLM (2025)	CVPR	Yes	SigLIP2	Qwen2.5-1.5B	T, V, L	-	-	-
InstructDoc (2024)	AAAI	Yes	CLIP	FlanT5, BLIP2	T, V, L	-	-	-
LayTokenLLM (2025b)	CVPR	Yes	-	LLaMA3-8B, Qwen1.5-7B	T, L	-	-	-
CoR (2026)	preprint	Yes*	Qwen2.5-VL	Qwen2.5-VL-7B	T, V	-	CoT, PE, HD	-
Retrieval-Augmented Pipelines								
AVIR (2025)	MMAAsia	No	Pix2Struct	Qwen2.5-VL-3B-AWQ	V	D	-	-
DFVC (2026)	Electronics	No	VisRAG, ColQwen	Qwen2-VL-2B-Instruct	V	D	-	-
DREAM (2025)	MM	No	InternVL2 ViT	InternVL2-4B	V	J	HD	-
M3DocRAG (2024)	preprint	No	ColPali, ColQwen	Multiple (Qwen/Intern/Idefics)	V	D	-	-
MM-R5 (2026)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B	V	D	CoT	-
MoLoRAG (2025b)	EMNLP	No	ColPali	Multiple (Qwen/DS/LLaVA)	V	D, Nav	-	1A
Self-Attention Scoring (2024)	ICDAR	No	Pix2Struct	Pix2Struct	V	D	-	-
SLEUTH (2025)	preprint	No	ColPali, Qwen3-VL ViT	Multiple (Qwen/GLM/Gemini)	V	D	HD	4A
SV-RAG (2025)	ICLR	No	InternVL2, Phi-3-vision	Multiple (Intern/Pali/Phi)	V	D	-	-
VDocRAG (2025)	CVPR	No*	Phi3V-4.2B	Phi3V-4.2B	V	D	-	-
CREAM (2024)	MM	Yes	Pix2Struct	LLaMA2-7B	T, V	J	HD	-
KGP (2024c)	AAAI	Yes	-	Multiple	T, L	S, IR, Nav	-	1A
MHier-RAG (2025)	preprint	Yes	Qwen-VL-Plus	Multiple (Qwen/DS/GPT)	T, V, L	J	CoT, HD	-
MLDocRAG (2026)	preprint	Yes	Qwen2.5-VL	Multiple (Qwen)	T, V, L	D	CoT, HD	-
MultiDocFusion (2025)	ACL	Yes	-	Multiple (Qwen/LLaMa/Mistral)	T, V, L	D	HD	-
M2RAG (2025a)	NeurIPS	Yes*	VisRAG-Ret	Multiple (Qwen)	T, V	J	-	3A
MDocAgent (2025)	preprint	Yes*	ColPali, ColQwen2	Multiple (Llama/Qwen)	T, V	J	-	5A
PDF-WuKong (2024)	preprint	Yes*	IXC2-VL-4KHD	IXC2-VL-4KHD	T, V	J	-	-
RAG-DocVQA (2025)	ICDAR	Yes*	DiT, Pix2Struct	Qwen2.5-VL-7B-Instruct	T, V, L	D	-	-
VisDoMRAG (2025)	NAACL	Yes*	ColPali / ColQwen2	Multiple (Qwen/Gemini/GPT)	T, V	J	CoT	3A, SC
Adaptive-Trajectory Pipelines								
Doc-V* (2026)	preprint	No	Qwen2.5-VL	Qwen2.5-VL-7B-Instruct	V	D, IR, QR, Nav	ReAct, HD	1A
DocReact (2025a)	ACL	No	ColPali, VisRAG	GPT-4o	V	D, IR, QR	ReAct, QD	1A
MACT (2025)	preprint	No	Multiple	Multiple (Qwen/MiMo/Intern)	V	D	CoT, PE	4A, SC, SV
VRAG-RL (2026)	NeurIPS	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B/7B-Instruct	V	D, IR, QR	ReAct, HD	1A
DocLens (2025a)	preprint	Yes	-	Multiple (GPT/Gemini/Claude)	T, V, L	J	CoT	4A, SA
DMAP (2026)	WWW	Yes*	ColPali	GPT-4o / Qwen-plus	T, V, L	J	HD	2A, SR
DocAgent (2025)	EMNLP	Yes*	-	Multiple (GPT/Gemini/Claude)	T, V, L	J, IR, Nav	ReAct, HD	3A
DocDancer (2026)	preprint	Yes*	Qwen3-VL-235B	Qwen3-4B/30B-A3B	T, V, L	J, IR, QR	ReAct	1A
MM-Doc-R1 (2026)	preprint	Yes*	Qwen2.5-VL ViT	Qwen3-4B/8B	T, V	S, IR, QR, Nav	ReAct, HD	3A
SimpleDoc (2025)	EMNLP	Yes*	ColQwen-2.5	Multiple (Qwen)	T, V	J, IR, QR	-	1A, SR, SV
ViDoRAG (2025a)	EMNLP	Yes*	NV-Embed-V2, ColQwen2	Multiple (Qwen/LLaMa/GPT)	T, V	J, IR	ReAct, HD	3A, SR, SV

1 Backbone

In the adaptive-trajectory family: a **general-purpose multimodal language model**, typically used as it is, with **no extra training or fine-tuning**.

2 Modalities

Encoding models combine **text, vision and layout**. Models in the second family mostly use **vision only**.

3 Search

Only appears once **retrieval is separated from answering**. Retrieval-augmented pipelines add a retrieval step.

Page-to-Document Encoding: Hi-VT5

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory

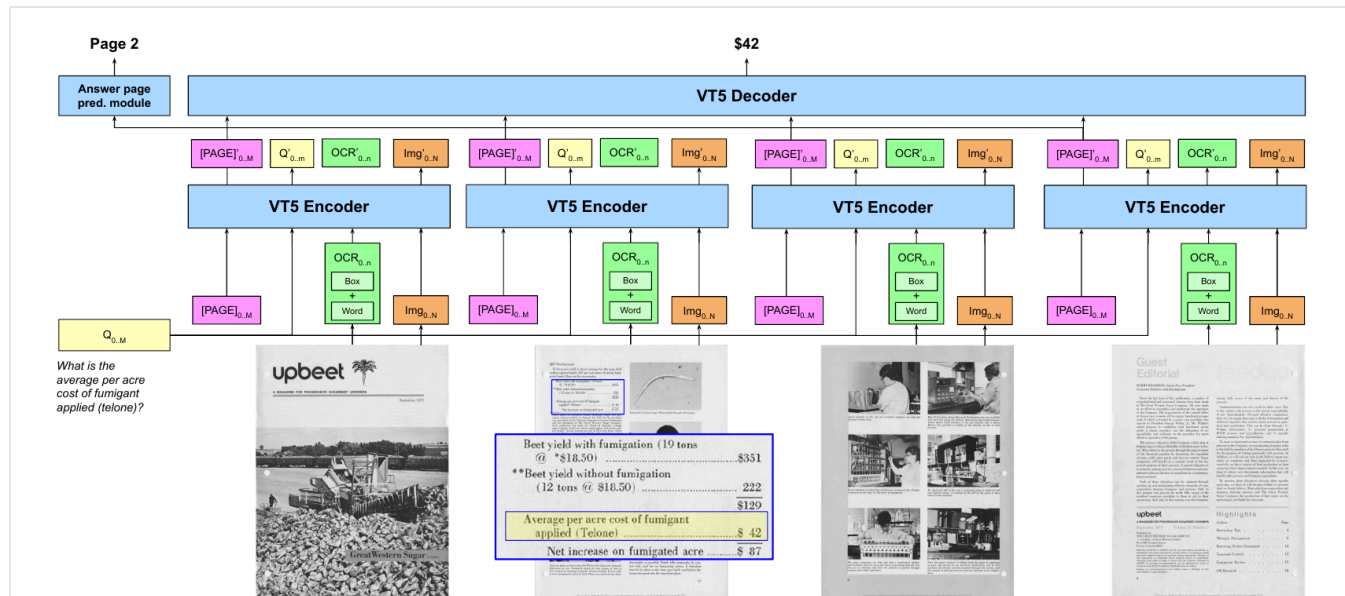


Figure from Tito et al., Pattern Recognition 2023

HOW IT WORKS

- Builds on **T5**, a text encoder-decoder, and adds visual and layout information.
- Each page is **encoded separately**, with the question, the recognised words and their positions, and patches of the page image.
- Ten learnable **[PAGE] tokens** summarise what on that page is relevant to the question.
- The decoder reads the **joined page tokens**, not the full pages, and generates the answer.

MLLM-Centric Adaptation: Docopilot

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory

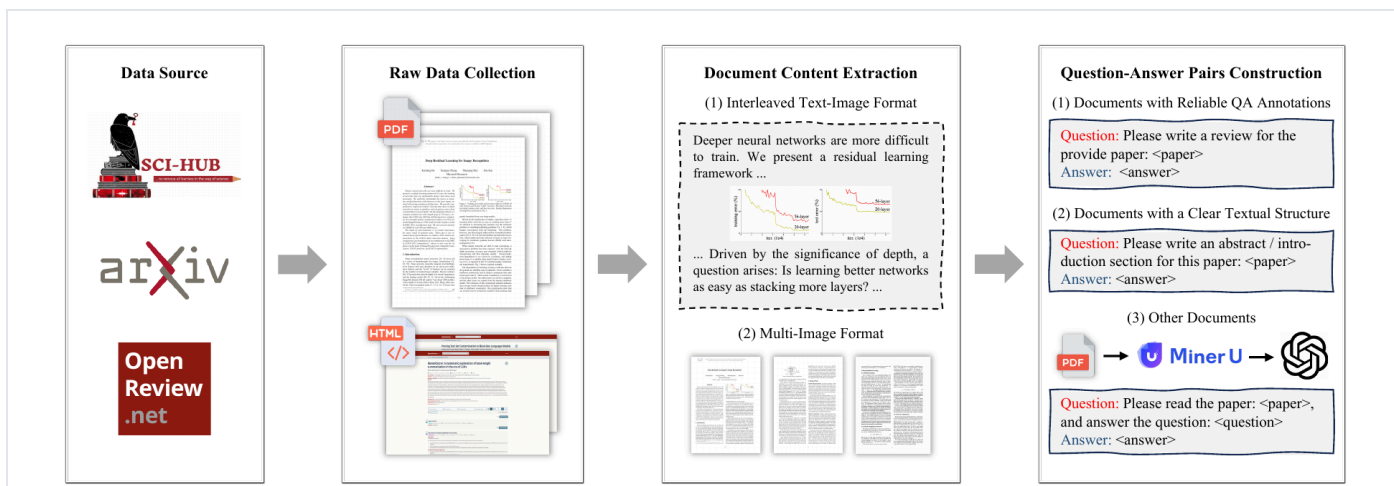


Figure from Duan et al., CVPR 2025

HOW IT WORKS

- Adapts a pretrained multimodal model to **read a multi-page document directly**, without retrieval.
- Teaches document-level dependencies through **better training data**: Doc-750K, about 750,000 question-answer pairs from scientific papers and reviews.
- **InternVL2** is fine-tuned on a data mixture that includes it.
- At inference, the page images and the question go into the model together, and it answers directly.

Retrieval-Augmented: MoLoRAG

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory

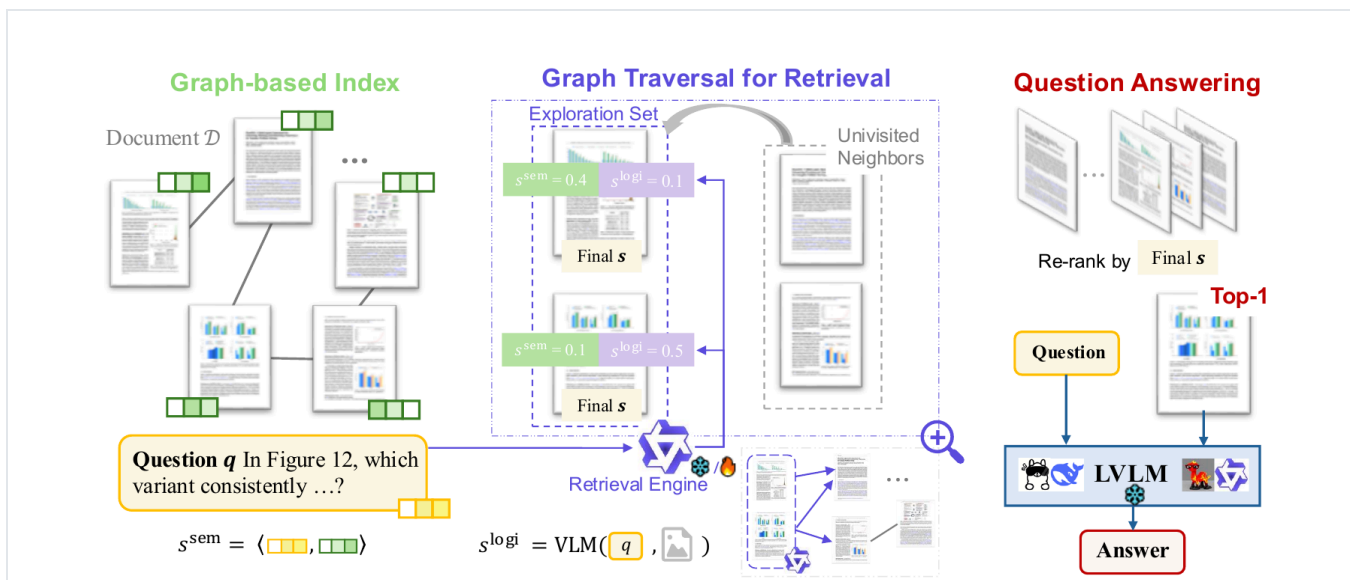


Figure from Wu et al., EMNLP 2025

HOW IT WORKS

- **Graph-based index.** A visual retriever represents each page, and pages are connected by their similarity.
- **Graph traversal.** The search starts from pages similar to the question. A small vision-language model helps score relevance, and the top K pages are kept.
- **Question answering.** The top K pages go to a separate vision-language model, the reasoner.

Adaptive-Trajectory: DocLens

Page-to-document encoding

MLLM-centric adaptation

Retrieval-augmented

Adaptive-trajectory

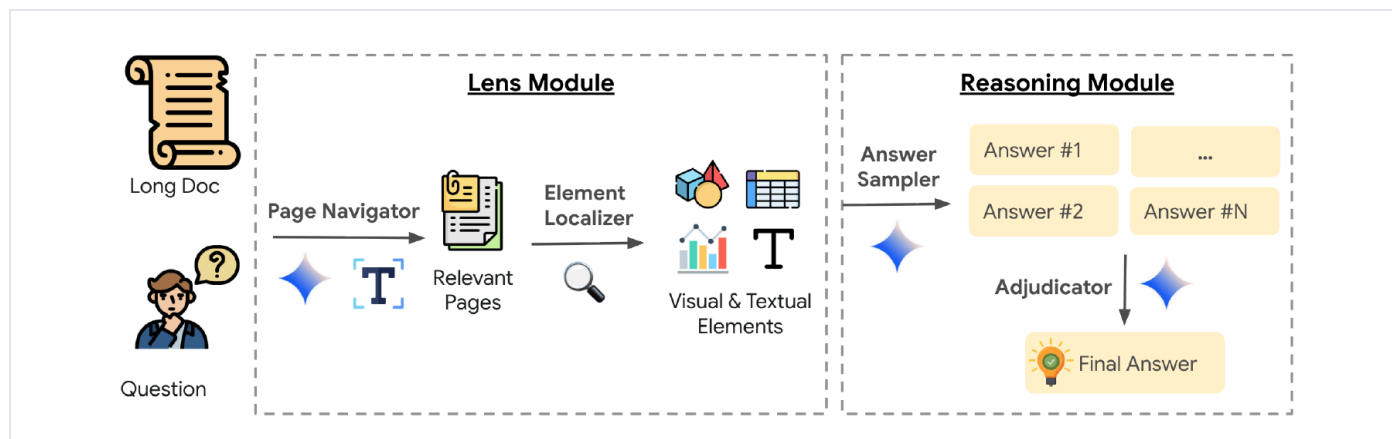


Figure from Zhu et al., ACL 2026

HOW IT WORKS

- A **tool-augmented multi-agent** framework, from Google and Peking University.
- **Lens module.** A page navigator uses page images and OCR text to find the relevant pages. An element localiser crops their figures, charts and tables.
- **Reasoning module.** Several candidate answers are sampled. An adjudicator compares their reasoning and synthesises the final answer.
- Find the pages, zoom in on their elements, compare candidate answers.

Table of Contents

- Introduction
- Survey of MP-VRDU

- **Empirical Attribution Study**

Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution

Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu

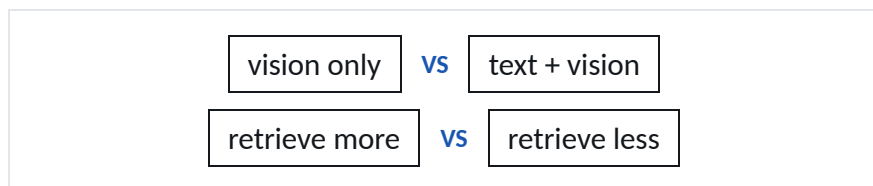
[Under review](#)

- Conclusion
- Proof of Concept: A Simple Method

Empirical Study: Research Gaps and Contributions

RESEARCH GAPS

- 1 **Competing design claims**, each tested on a different pipeline and dataset.



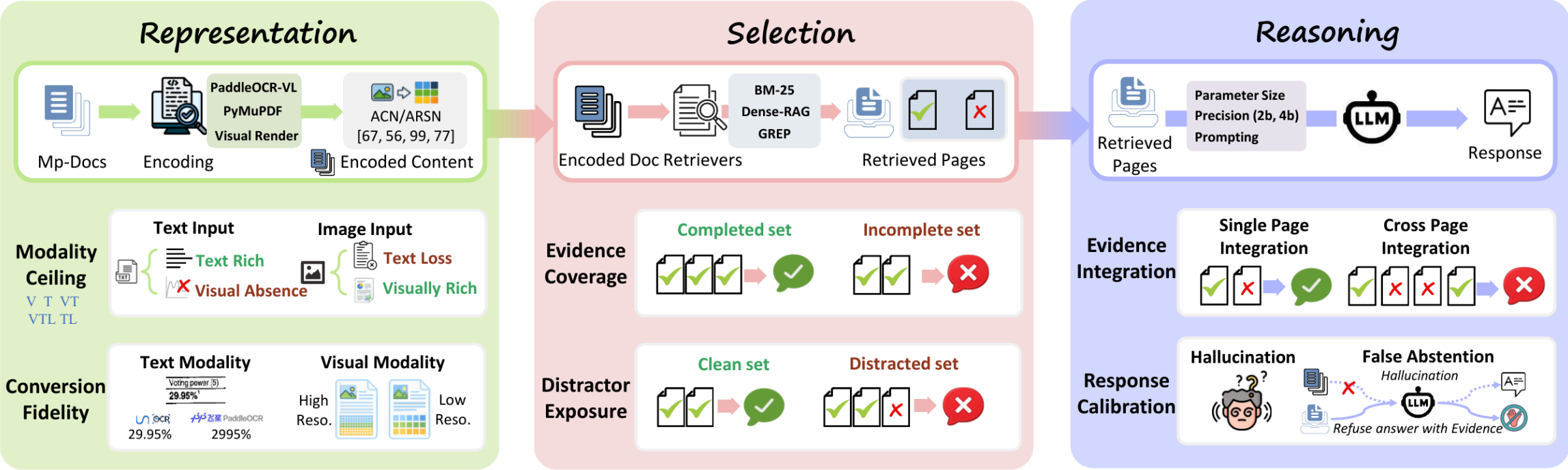
- 2 **No end-to-end empirical study** of where document understanding systems fail.

CONTRIBUTIONS

- 1 A **unified view of failure**: three architecture-independent loci, representation, selection and reasoning.
- 2 Controlled interventions in **one pipeline**, across document collections and independently developed model families.
- 3 **Practical guidance** for designing and deploying systems **under a compute budget**, from qualitative examples and accuracy-cost comparisons.

This is the local-deployment setting from the constraints slide.

Attribution Framework



REPRESENTATION

Was the information preserved when the page was converted?

SELECTION

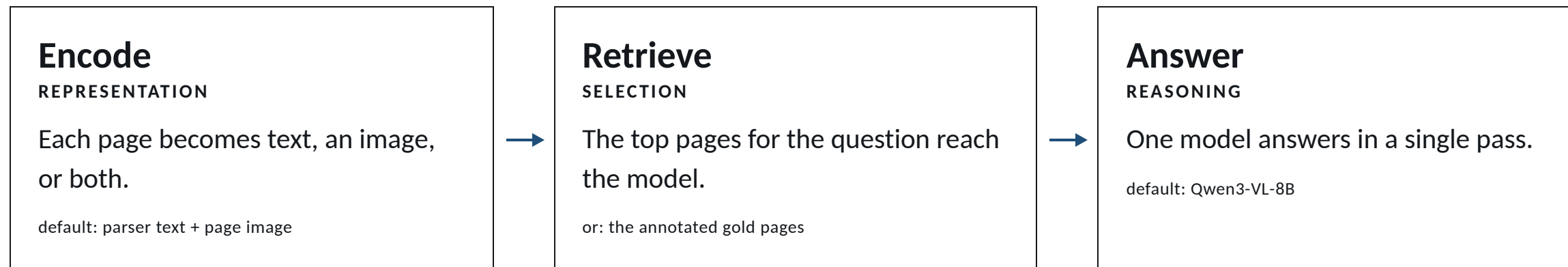
Did it actually reach the model?

REASONING

Did the model use it correctly?

Attribution by Construction

One simple pipeline. **Change one stage, hold the other two fixed.**



- **Gold pages** bypass retrieval, so selection can be taken out of the picture.
- **MMLongBench-Doc**: 135 documents, annotated with evidence pages, evidence type and answerability.

Representation: Results

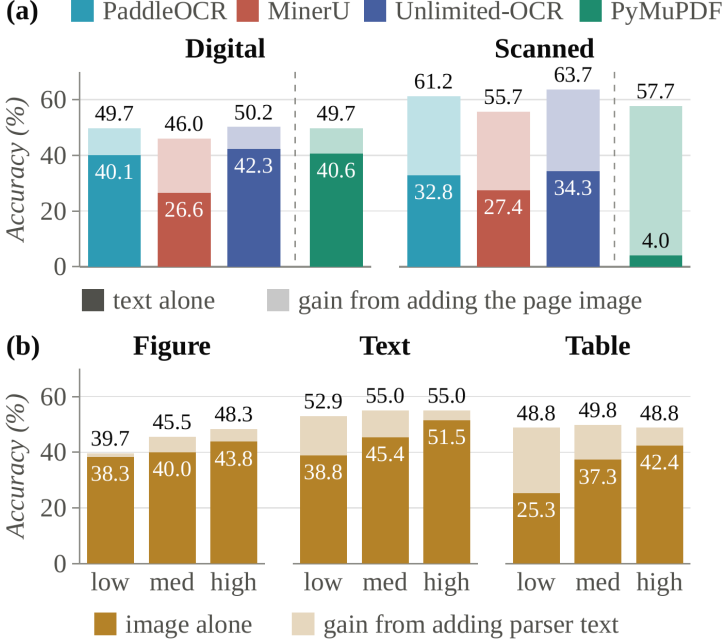
Is the evidence in the input?

Modality ceiling

Evidence	Abstention				Accuracy			
	T	TL	TLV	V	T	TL	TLV	V
MMLongBench-Doc								
Text	23.0	11.5	7.3	8.0	36.9	44.4	51.0	42.3
Layout	20.3	10.2	0.8	4.2	26.3	33.9	49.2	43.2
Table	15.9	7.2	6.7	9.1	35.6	44.7	44.7	34.1
Chart	32.6	28.1	7.3	6.2	16.3	16.9	42.7	40.4
Figure	36.6	24.8	7.6	7.9	14.5	22.1	42.4	39.0
All	26.3	17.2	6.2	7.1	29.0	35.7	49.5	42.6
LongDocURL								
Text	4.6	3.5	1.5	1.6	71.8	75.6	78.2	70.0
Layout	2.1	1.6	0.7	0.9	45.0	49.2	51.3	49.3
Table	5.0	3.2	1.1	0.8	57.7	65.6	63.6	52.4
Figure	6.7	8.6	1.5	1.3	59.9	62.8	69.9	61.9
All	4.7	4.3	1.3	1.3	60.3	64.5	67.3	59.4

Text-only input refuses **chart and figure** questions most. Page images bring refusal **under 10%**, yet images alone trail text plus images.

Conversion fidelity



Adding page images cuts the spread across parsers from **15.7 to 4.2 points**. Parser text makes resolution matter less.

Representation: Design and Deployment

Is the evidence in the input?

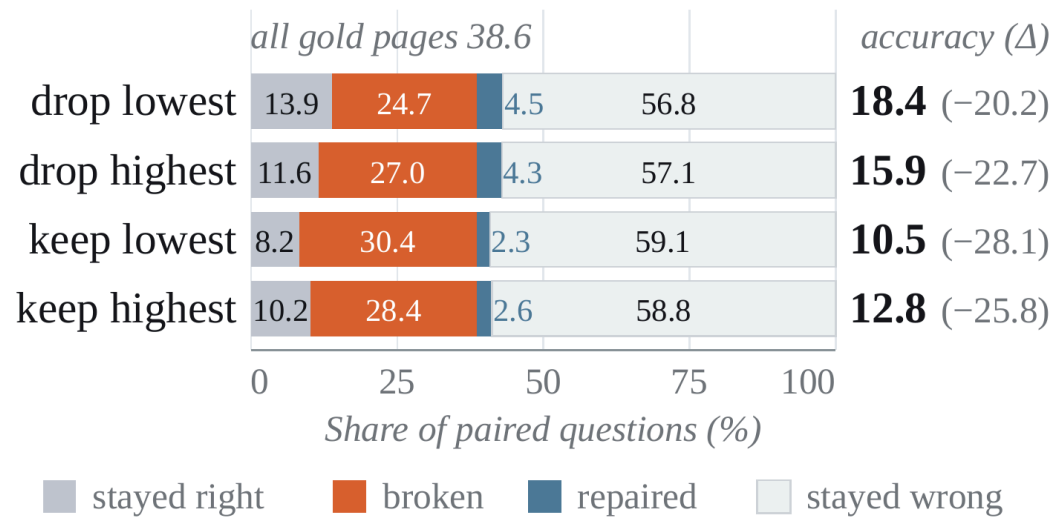
Modality ceiling	two inline icons, parsed as 	Conversion fidelity	scanned slide, text by parser
Q What are the colors of the icons that the users touch to move the app up and remove the app respectively? Gold gray, red T ✗ "The document does not specify the colors" TL ✗ "it does not mention any visual details" --- ▼ add the page image ----- V ✓ "The icon is gray. ... The icon is red." TLV ✓ "gray (a gray upward arrow) ... red" Colour exists only in pixels; a text input leaves the reasoner nothing to read.		Q What is the name of the company that created this presentation? Gold AVALAUNCH MEDIA PaddleOCR ✗ "Asterks" (parse: KICK ASTERKS) MinerU ✗ "does not provide any evidence" --- ▼ re-convert, or add the image ----- Unlimited ✓ "AVALAUNCH MEDIA" with image ✓ "Avalaunch Media ... in the banner" The error sits in the transcript, not the reasoner, and each parser fails differently.	

- | | |
|------------|---|
| DESIGN | — Complementary modalities recover what one input loses: prioritise reliable embedded text, add images when appearance matters or extraction is unreliable, re-parse when the text is inadequate. |
| DEPLOYMENT | — Embedded text plus images matches parser text plus images (50.1% against 49.5%) with fewer tokens. Images cost 26.3 s of prefill against 1.8 s, so set modality and selection depth together. |

Selection: Results

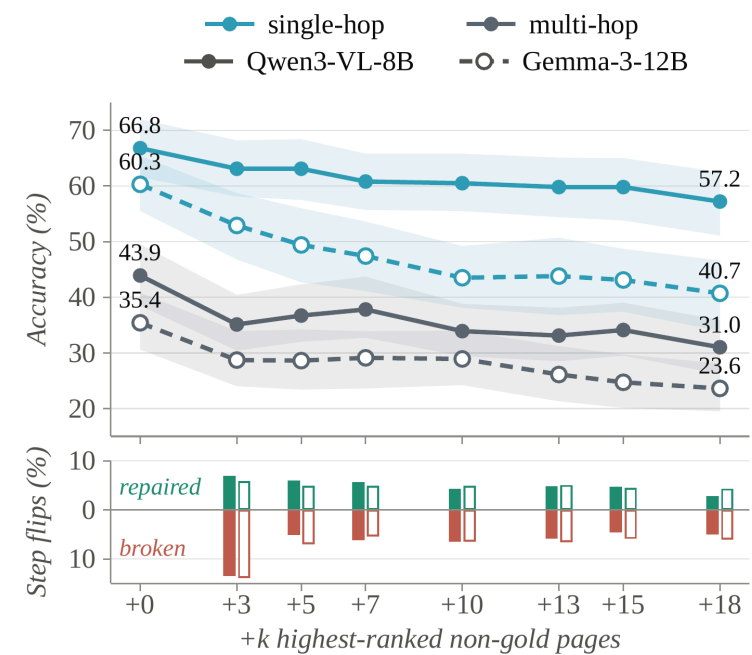
Is the right evidence in context?

Evidence coverage



Removing **one gold page** takes accuracy from **38.6% to 18.4%** or 15.9%. Retrieval rank does not predict which page is safe to lose.

Distractor exposure



With every gold page kept, three extra pages drop multi-hop accuracy from **43.9% to 35.1%**. Both backbones show it.

Selection: Design and Deployment

Is the right evidence in context?

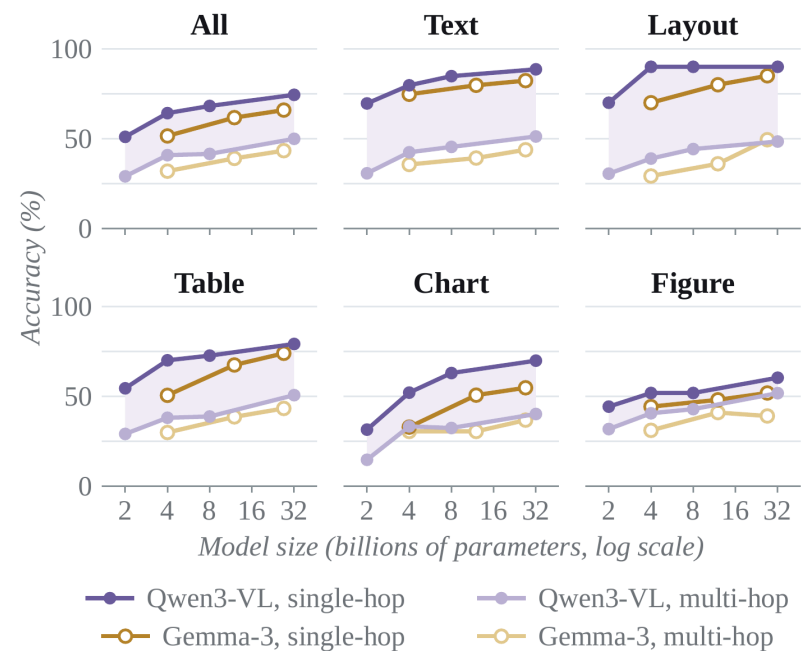
Evidence coverage	author list on pages 13, 14	Distractor exposure	10 gold pages of a 20-page deck
Q How many chapter authors of this report come from Columbia University? Gold 3 page 13 ❌ “two” (Horton, Kinney) page 14 ❌ “one” (Garcia-Pando) --- ▼ add the complementary page ----- both pages ✅ “three” (Horton, Kinney, Garcia-Pando) Each page is relevant, yet neither suffices: the answer is a property of the page set.		Q How many lateral bar graphs are present in the document? Gold 10 +3 pages ❌ “there are 12 lateral bar graphs” +5 pages ❌ “there are 13 lateral bar graphs” --- ▼ remove competing pages, or read on ----- gold pages ✅ “there are 10 lateral bar graphs” +7 to +18 ✅ “there are 10 lateral bar graphs” A few added pages inflate the count; the gold set, or more context, counts right.	

DESIGN	— Prioritise collective evidence coverage while controlling distractors: structured navigation, summary-based reranking, or iterative retrieval.
DEPLOYMENT	— ColQwen3 lifts recall at 5 from 58.1% to 81.0% over BM25, but needs a GPU and 110 ms per query against 0.3 ms. Judge retrieval depth by end-to-end answers and latency , not recall alone.

Reasoning: Results

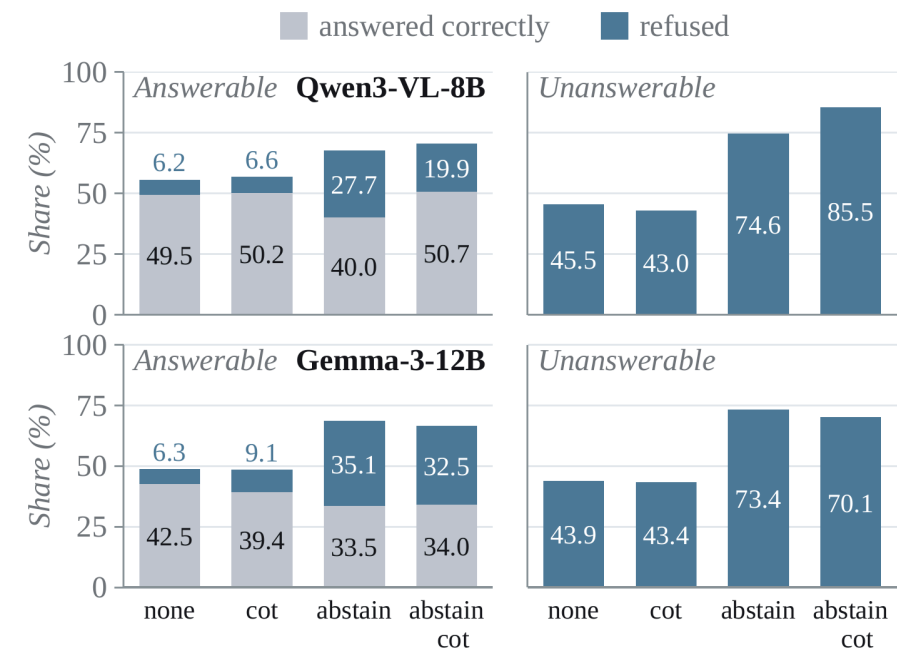
Is the evidence used correctly?

Evidence integration



From 2B to 32B, multi-hop accuracy rises from **29.0%** to **49.9%**, but the gap to single-hop stays.

Response calibration



An abstain instruction lifts correct refusal to 74.6%, but false refusal goes from **6.2%** to **27.7%**. Reasoning first recovers part of it.

Reasoning: Design and Deployment

Is the evidence used correctly?

Evidence integration

a diagram, an ablation table

Q If I drop the green module shown in Figure 6, what is the absolute percentage drop on 100-shot TACREV setting?

Gold 0.5%

2B ❌ "2.4%" (reads another table)

4B ❌ "-9.6%" (right table, wrong row)

8B ❌ " $(75.9 - 74.9) / 75.9 \dots \approx 1.32\%$ "

--- ▼ **a larger reasoner** -----

32B ✅ " $75.9 - 75.4 = 0.5$ "

Each size breaks a different step of one chain: locate the table, read the cell, compute.

Response calibration

two tables, two pages

Q How much lower is the accuracy of GPT-4 on SituatedQA compared to MMLU Chemistry?

Gold 16.7%

abstain ❌ "Not answerable" (false refusal)

--- ▼ **reason before deciding** -----

abstain_cot ✅ " $79.9\% - 63.2\% = 16.7\%$ "

none ✅ "... 16.7 percentage points lower"

cot ✅ "Step 1: ... Answer: 16.7%"

The question is answerable; the abstain instruction turns multi-step difficulty into refusal.

DESIGN

- **Match capacity and inference strategy to the operation.** Separating answer construction from verification helps, at the cost of extra computation.

DEPLOYMENT

- Quantisation keeps accuracy at a **smaller footprint but is not faster**: under a memory limit prefer a quantised larger model, under a latency limit fewer reasoning steps or a faster backbone.

Conclusion

SURVEY

Organise

Multi-page document understanding is a problem of **evidence management**.

EMPIRICAL STUDY

Locate

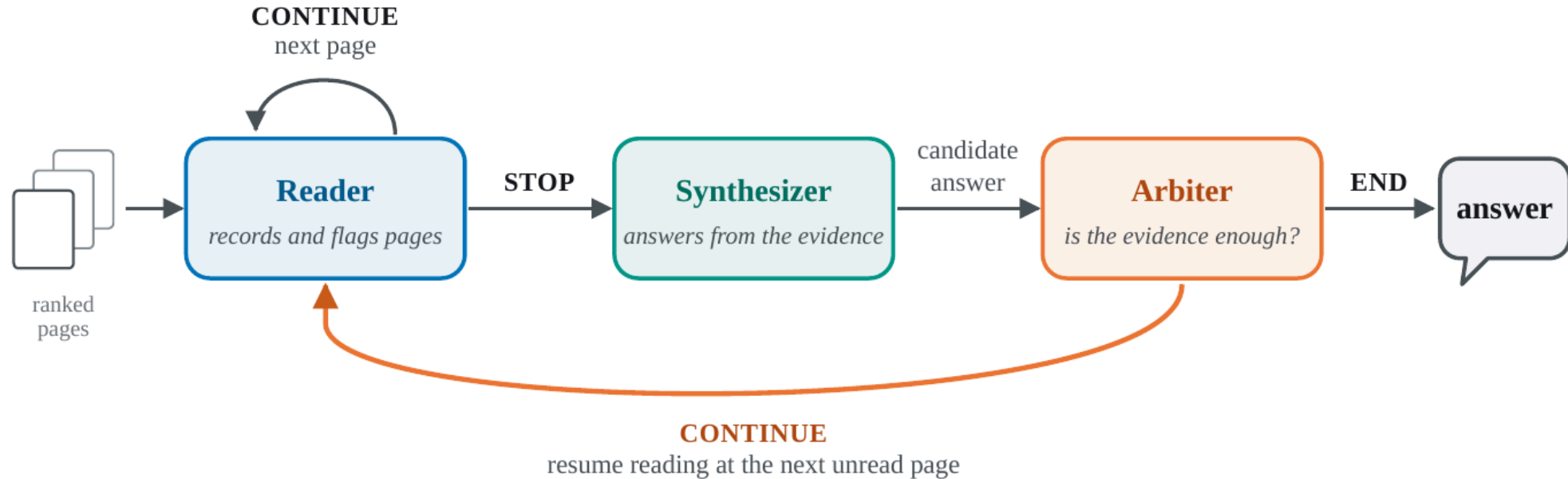
Failures sit at **representation, selection, or reasoning**, and each needs a different fix.

Thank you Questions?

Table of Contents

- Introduction
- Survey of MP-VRDU
- Empirical Attribution Study
- Conclusion
- **Proof of Concept: A Simple Method**

From Findings to a Simple Method



REPRESENTATION

Every page is read as **text and image together**.

SELECTION

Read broadly, then pass on compact records, not full pages.

REASONING

A separate arbiter decides **whether the evidence is enough**.

Simple Method Results

Method	OCR	Trained	MMLongBench-Doc				LongDocURL			
			SIN	MUL	UNA	ACC	UND	REA	LOC	ACC
Gold pages + VLM	✓	×	57.7	37.4	52.5	49.5	69.7	63.6	59.5	65.7
Whole document (via DocTrace)	×	×	46.9	35.3	25.8	38.5	–	–	–	45.1
M3DocRAG (via SLEUTH)	×	×	–	–	48.6	45.9 ^a	60.1	52.4	45.5	54.6
DocAgent (via MARDoc)	✓	×	–	–	–	46.6	–	–	–	–
MDocAgent (via SLEUTH)	✓	×	–	–	49.7	47.8 ^a	57.5	51.5	44.7	53.1
MoLoRAG (via SLEUTH)	×	×	–	–	51.7	48.8 ^a	63.4	52.7	49.3	57.6
SLEUTH	×	×	–	–	67.4	52.8 ^a	65.7	53.0	53.6	60.0
DocTrace	✓	✓	53.2	41.1	70.5	52.9	–	–	–	56.4
Ours (4-bit)	✓	×	60.5	39.2	76.9	56.5	67.2	66.3	47.9	61.3

- **56.5%** on MMLongBench-Doc and **61.3%** on LongDocURL: highest among the Qwen3-VL-8B systems, training-free, at 4-bit.
- Multi-page questions still trail DocTrace, **39.2 against 41.1**.

Inference-Time and Training Strategies

Three evidence-management decisions. Each can be **prompted around a frozen MLLM**, or **trained into it**.

Retrieval and navigation

TRAINING-FREE

Similarity-based or relation-aware retrieval

TRAINED

Modality-alignment, cross-page-fusion, logical-relevance

Reasoning strategies

TRAINING-FREE

Chain-of-Thought, Self-Reflection, Self-Consistency

TRAINED

Trace-imitation, then reward-driven (GRPO)

Agentic methods

TRAINING-FREE

ReAct and tool use, multi-agent orchestration

TRAINED

Trajectory-distillation, reinforcement-learned

Retrieval and Navigation

Retrieval **narrows a long document to compact evidence units** (chunks, pages, or graph nodes) for downstream reasoning.

- **Similarity-based retrieval.** Scores each evidence unit independently against the query and returns the top-ranked subset. Sparse lexical signals such as BM25 are the baseline; dense retrieval uses vision-language late-interaction encoders such as ColPali and ColQwen, often with an LLM reranking stage after.
- **Relation-aware retrieval.** Retrieves along explicit document structure. Tree-based variants follow section hierarchies and outlines, so chunks keep their parent context; graph-based variants traverse typed edges between headings, entities, and cross-references.
- **The trade-off.** Independent scoring handles paraphrase and cross-modal queries, but cannot follow cross-page references or multi-hop dependencies. Relation-aware methods can, at the cost of parser dependence and traversal cost.

Evidence missed or poorly ranked at retrieval is **unrecoverable at generation**. What is retrieved still has to be reasoned over.

Reasoning Strategies

Reasoning strategies act over a **single evidence buffer assembled before the reasoning call**: the full document representation, a retrieved subset, or evidence from an agentic step. None updates model parameters.

- **Sequential reasoning: Chain-of-Thought.** Extends a single reasoning thread. CoT binds evidence from different pages into one chain and decomposes multi-hop questions into single-hop steps.
- **Sequential reasoning: Self-Reflection.** The model evaluates its own draft answer and revises it across rounds, optionally against a verifier or critic.
- **Parallel exploration.** Several reasoning paths over the same buffer. Self-Consistency samples multiple chains and takes the majority answer; sampling-adjudication replaces the vote with an LLM judge. Cost scales with the number of samples. Tree-of-Thoughts has seen little uptake.

A chain can only reason over the evidence already assembled. It **cannot recover what retrieval missed**.

Agentic Methods

Agentic methods put one or more **LLM controllers** in charge: either iterating evidence acquisition under inference-time control, or coordinating across specialised roles.

- **ReAct and tool-augmented reasoning.** A single controller interleaves thought, action, and observation. Actions are document-side tool calls (retrievers, parsers, OCR engines) or structured navigation over a pre-built document representation.
- **Multi-agent orchestration.** Modality-specialised decomposition gives text and visual evidence separate controllers plus a synthesis controller. Role-specialised decomposition splits planning, execution, verification, and synthesis.
- **What the adaptive trajectory buys.** Compute scales with question difficulty rather than document length, the single-step recall ceiling of retrieval-augmented pipelines is lifted, and the action sequence is an inspectable audit trail.

The cost is **controllability**: inference cost is hard to bound, and a controller can commit early to wrong evidence and continue down the wrong trajectory.

Training Strategies

Most systems adopt a backbone **already pretrained on single-page data**, so fine-tuning carries the weight. Training internalises the inference-time evidence-management decisions into model parameters.

Trained retrieval

Modality-alignment aligns visual page embeddings with OCR-derived text. **Cross-page-fusion** adds neighbouring-page context to each page representation. **Logical-relevance** ranks pages by their position in a cross-page reasoning path.

RETRIEVER

Trained reasoning

Trace-imitation supervises the backbone on demonstration chains in a fixed schema. **Reward-driven** training then refines it with GRPO, rewarding answer accuracy, evidence-page recall, and format compliance.

REASONER

Trained agentic methods

Trajectory-distillation clones a stronger teacher's think-action-observation turns through SFT. **Reinforcement-learned** variants optimise the policy directly against trajectory-level rewards.

CONTROLLER

Two techniques recur across all three: **parameter-efficient tuning** (LoRA, adapters) and **multi-stage curricula** that go from single-page to document-scale supervision. All of it depends on the available datasets.

Datasets

Datasets serve three roles, and the multi-page setting makes all three harder: **per-page annotation does not scale** to documents of tens or hundreds of pages, and long PDFs are harder to source and license than single pages.

Pretraining

The exception rather than the norm. Confined to a few page-to-document and MLLM-centric systems that do continued pretraining on an existing backbone, because multi-page corpora are still small.

RARE

Fine-tuning

Has converged on LVLM-synthesised supervision over scraped PDFs, with reinforcement learning increasingly layered on top to shape evidence-grounding and trajectory behaviour.

LVLM-SYNTHESISED

Benchmarking

Three tiers: single-page extractive baselines, mid-length multi-page suites, and long-document multimodal benchmarks that need cross-page integration. The same small pool of PDFs gets reused.

CONTAMINATION RISK

Answer-only accuracy masks retrieval failure and rewards correct answers reached through incorrect reasoning. Newer benchmarks score retrieval and page-localisation as well, but it is not yet standard.

Datasets and Evaluation

PRIMARY

MMLongBench-Doc

135	48.3	847	244
documents	mean pages	answerable	unanswerable

- Annotated with **evidence pages, evidence sources and answerability**, which the experiments need.
- 367 of the answerable questions need two or more pages.

COMPLEMENTARY

LongDocURL

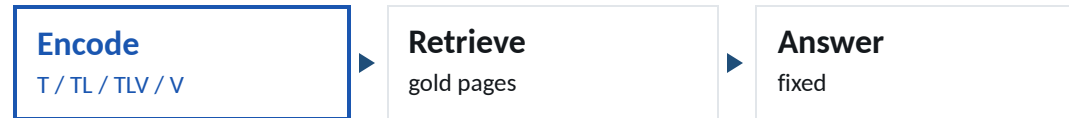
396	85.6	2,325
documents	mean pages	questions

- **Born-digital only**, so answers are often in the text layer, even for figure questions.
- Used for the modality-ceiling result only: its evidence pages are not a minimal set.

Judge: Gemini-2.5-Flash at temperature zero returns correct, incorrect or abstained. Agreement is **96.0%** with GPT-4o and **98.3%** with a human annotator.

Modality Ceiling

Evidence	Abstention				Accuracy			
	T	TL	TLV	V	T	TL	TLV	V
MMLongBench-Doc								
Text	23.0	11.5	7.3	8.0	36.9	44.4	51.0	42.3
Layout	20.3	10.2	0.8	4.2	26.3	33.9	49.2	43.2
Table	15.9	7.2	6.7	9.1	35.6	44.7	44.7	34.1
Chart	32.6	28.1	7.3	6.2	16.3	16.9	42.7	40.4
Figure	36.6	24.8	7.6	7.9	14.5	22.1	42.4	39.0
<i>All</i>	26.3	17.2	6.2	7.1	29.0	35.7	49.5	42.6
LongDocURL								
Text	4.6	3.5	1.5	1.6	71.8	75.6	78.2	70.0
Layout	2.1	1.6	0.7	0.9	45.0	49.2	51.3	49.3
Table	5.0	3.2	1.1	0.8	57.7	65.6	63.6	52.4
Figure	6.7	8.6	1.5	1.3	59.9	62.8	69.9	61.9
<i>All</i>	4.7	4.3	1.3	1.3	60.3	64.5	67.3	59.4



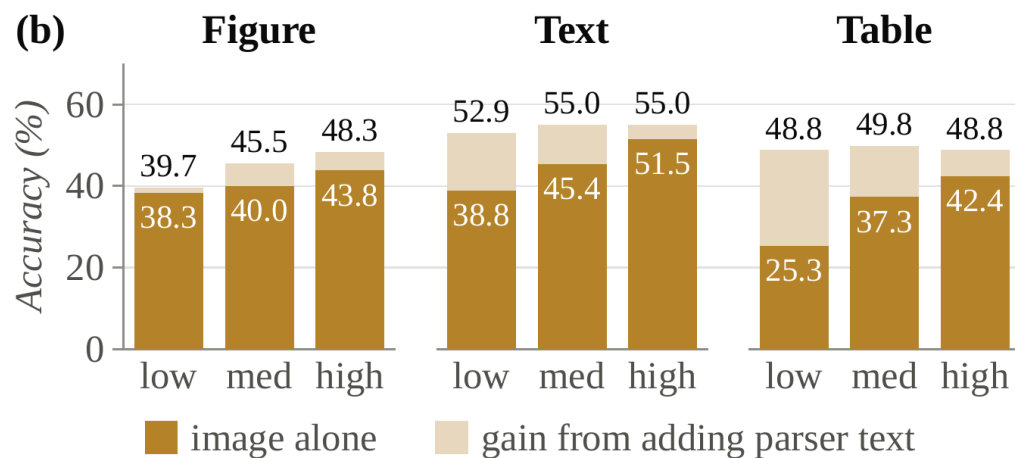
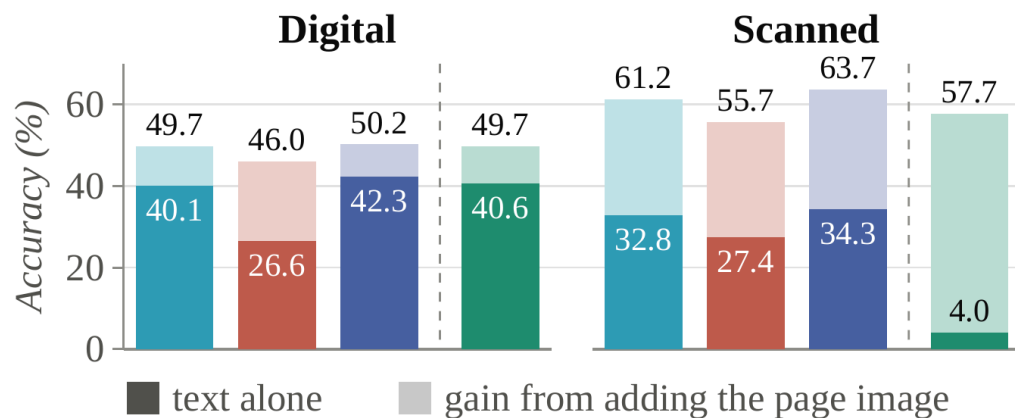
REPRESENTATION

Evidence accessibility depends on modality

- On MMLongBench-Doc, text-only input refuses **chart and figure** questions most. Page images bring refusal **under 10%**.
- Images alone stay below text plus images: 42.6% against 49.5%.
- On born-digital LongDocURL, adding vision to parser text gives only **2.8 points** overall, 7.1 on figure questions.

Conversion Fidelity

(a) PaddleOCR MinerU Unlimited-OCR PyMuPDF



Encode
parser, resolution

Retrieve
gold pages

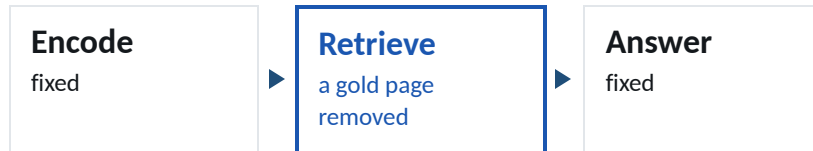
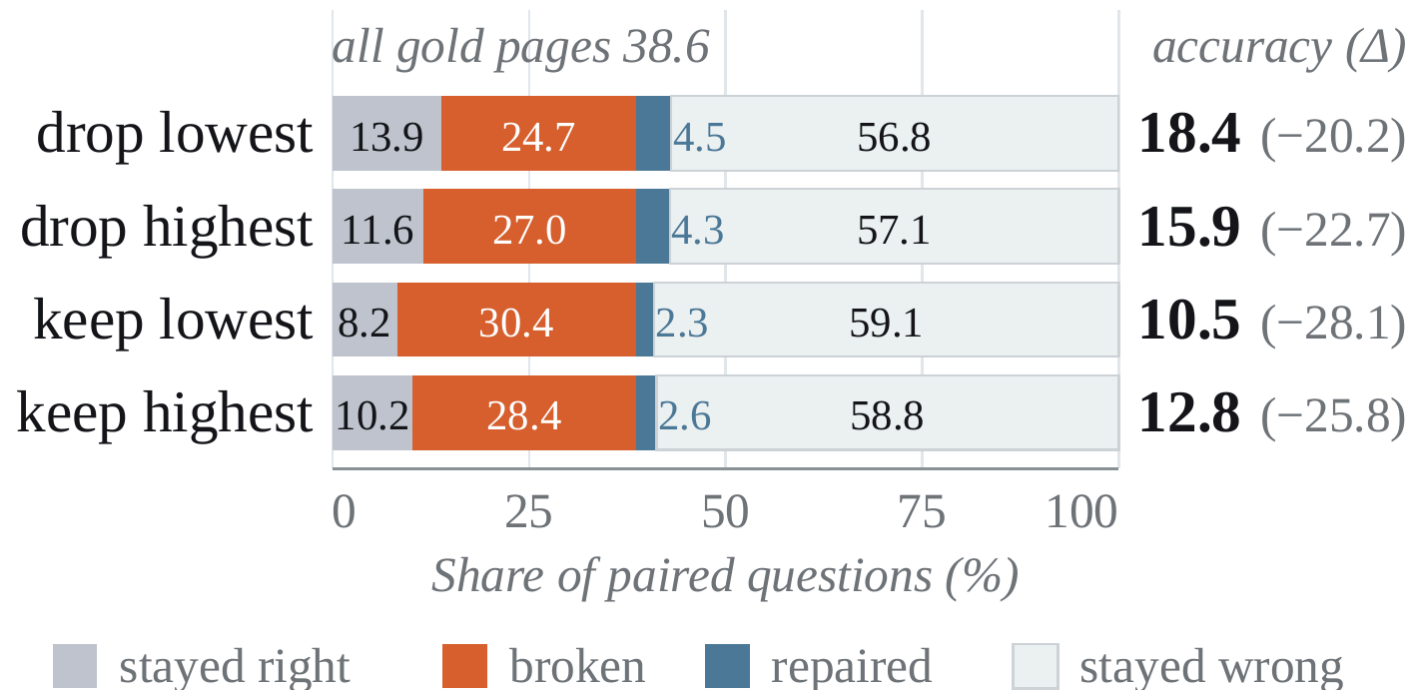
Answer
fixed

REPRESENTATION

Complementary modalities reduce conversion sensitivity

- Adding page images cuts the spread across parsers from **15.7 to 4.2 points** on digital documents.
- Higher resolution helps images alone far more than text plus images.
- Scanned documents stay sensitive to the parser, and figure questions to resolution.

Evidence Coverage

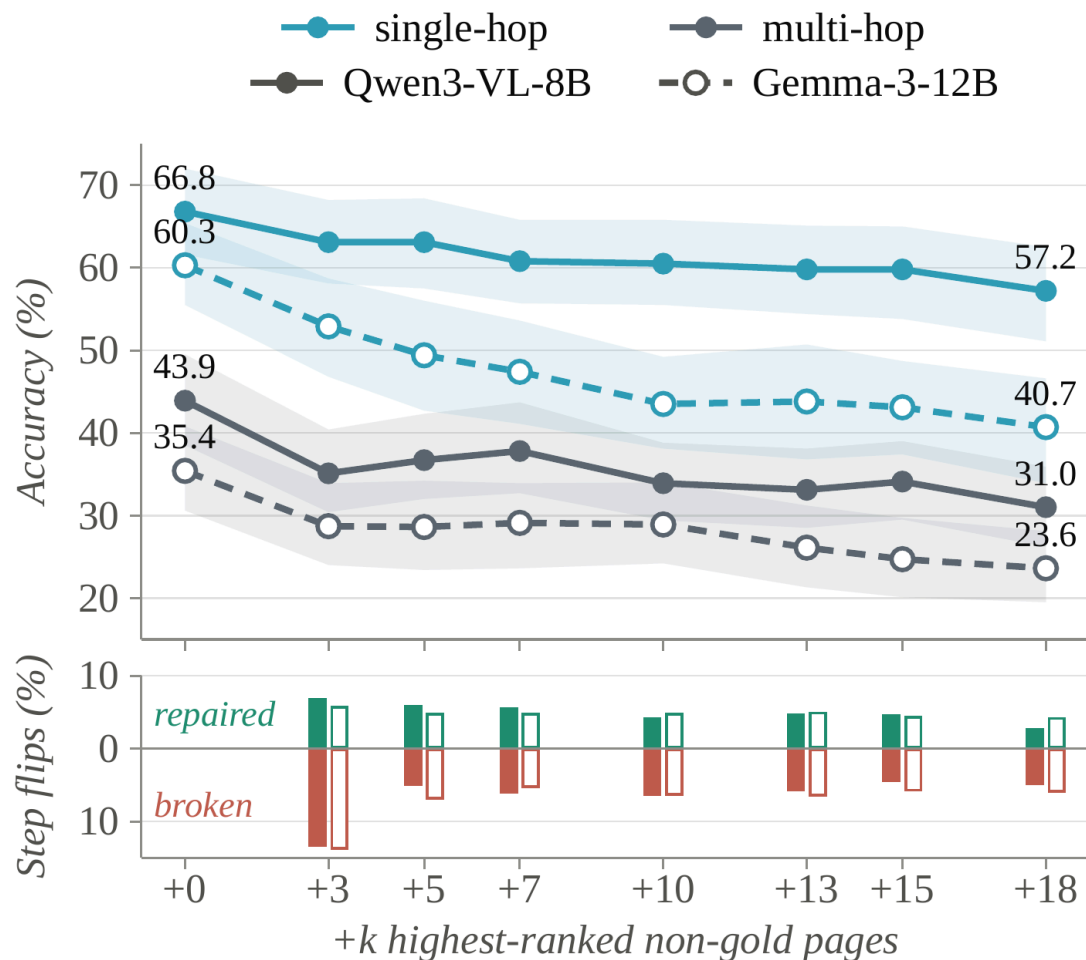


SELECTION

Omission causes large accuracy losses

- Removing **one gold page** takes accuracy from **38.6% to 18.4%** (lowest-ranked) or 15.9% (highest-ranked).
- Correct answers break far more often than wrong ones are repaired.
- Retrieval rank does not predict which page is safe to lose.

Distractor Exposure



Encode
fixed

Retrieve
gold + k other pages

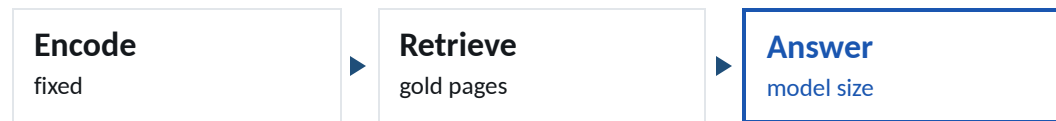
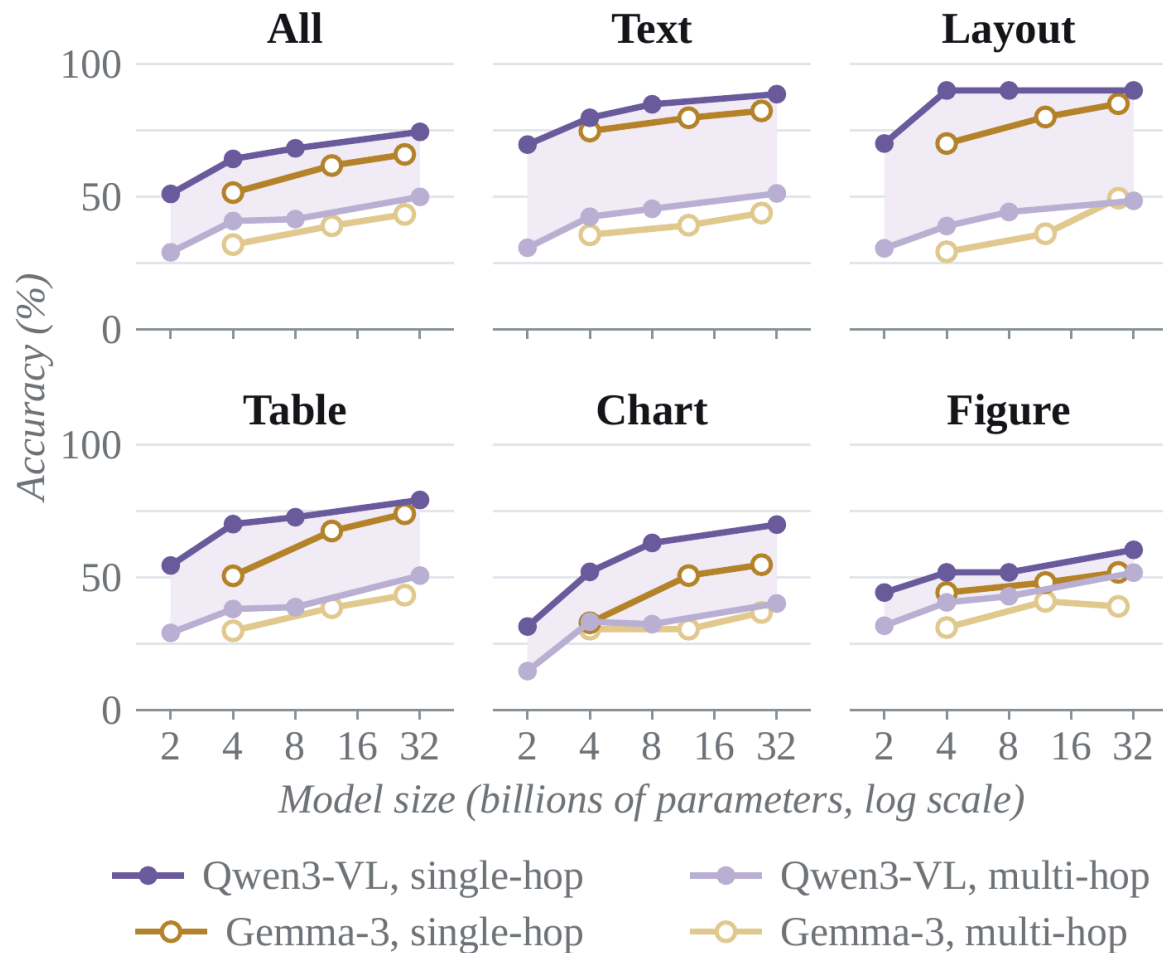
Answer
fixed

SELECTION

Multi-hop accuracy drops sharply with added context

- Qwen3-VL-8B: multi-hop falls from **43.9% to 35.1%** after three extra pages. Single-hop declines gradually, 66.8% to 57.2%.
- Gemma-3-12B shows the same trend: multi-hop 35.4% to 23.6%, single-hop 60.3% to 40.7%.
- Every gold page is still there. Extra context both breaks and repairs answers.

Evidence Integration

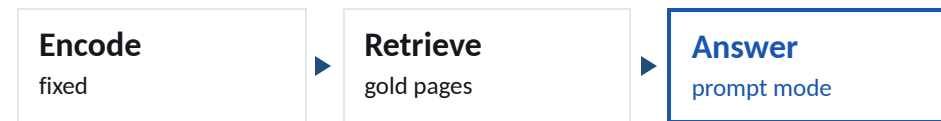
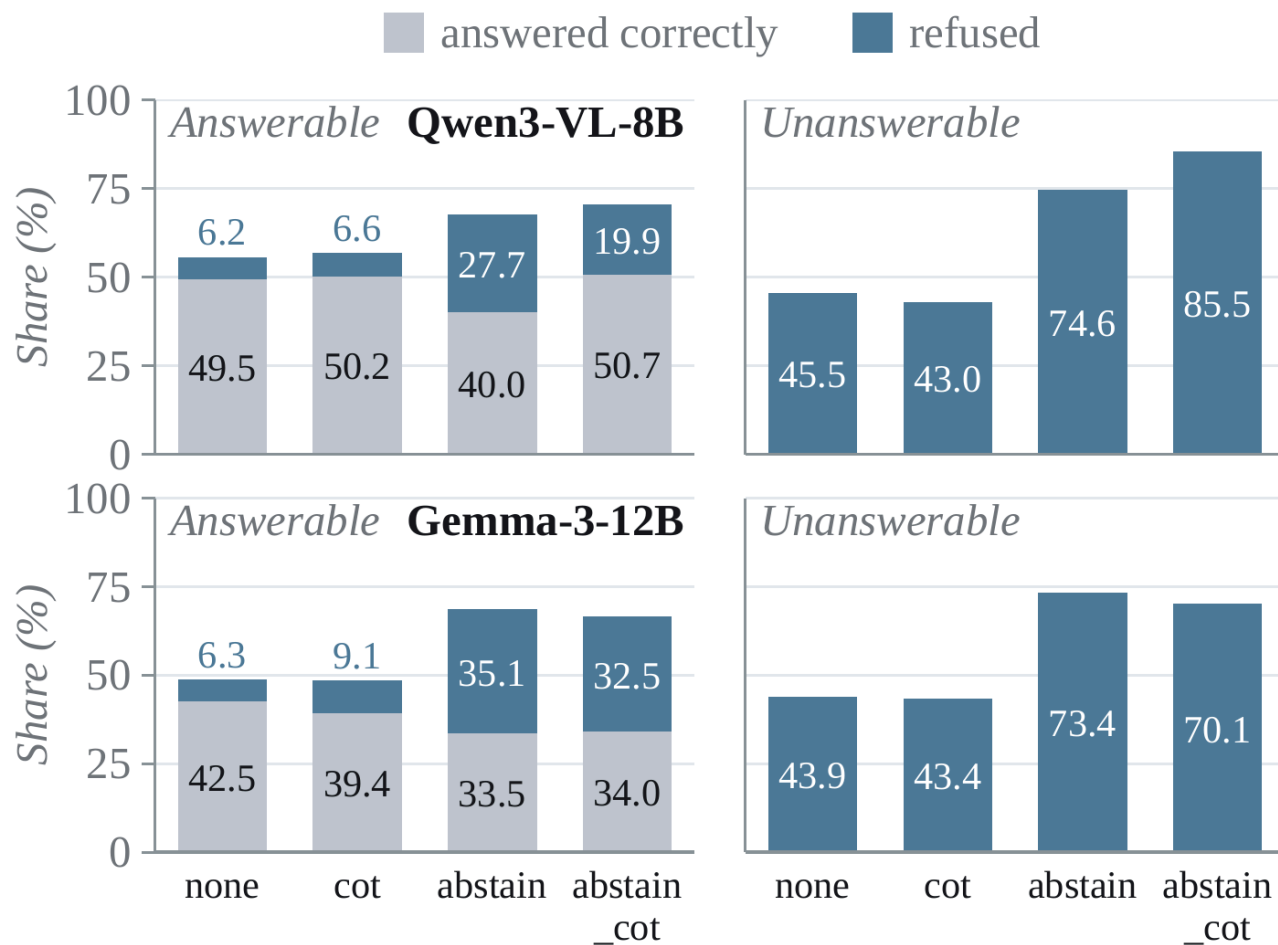


REASONING

Scaling helps, but the multi-hop gap persists

- Qwen3-VL from 2B to 32B: multi-hop rises from **29.0% to 49.9%**, single-hop from 51.0% to 74.4%.
- The gap between them stays roughly the same.
- Gemma-3 shows the same separation.

Response Calibration



REASONING

Reasoning before abstaining limits false refusal

- Qwen3-VL-8B, abstain: unanswerable refusal **45.5% to 74.6%**, but false refusal **6.2% to 27.7%** and accuracy 49.5% to 40.0%.
- Adding CoT: false refusal 19.9%, accuracy 50.7%, unanswerable refusal 85.5%.
- Gemma-3-12B gains much less from CoT: false refusal 35.1% to 32.5%.

Deployment: Accuracy and Cost

Representation	Acc.	Text	Vis.	Tokens	Prefill	Decode
T	29.0	499	0	499	1.4	5.7
TL	35.7	874	0	874	1.8	6.6
TV	50.1	500	459	959	–	–
TLV	49.5	874	459	1333	26.3	6.7
V	42.6	20	459	479	29.0	7.1
Retriever	R@1	R@5	R@10	Idx. (s)	Qry. (ms)	GB
BM25	32.7	58.1	68.1	0.02	0.3	CPU
BGE-M3	32.8	61.3	72.4	0.01	47	1.2
Qwen3-Emb.	33.5	63.9	74.8	0.19	89	13.1
ColModernVBERT	47.1	74.0	82.7	0.91	41	5.1
ColQwen2.5	49.2	77.2	84.8	1.50	293	8.6
ColQwen3	53.7	81.0	87.9	0.57	110	9.4
Reasoner	T	TL	TLV	V	GB	Time (s)
Qwen3-VL-8B, bf16	31.9	38.4	52.4	45.5	17.5	30.7
8-bit	32.5	37.9	53.6	45.7	10.3~	43.5
4-bit	31.8	39.4	51.8	45.5	6.8~	35.4
Qwen3-VL-32B, 4-bit	37.0	42.1	59.1	52.3	20.0~	47.5
InternVL3-8B	19.3	25.3	32.6	24.1	15.9	5.3
MiniCPM-V-4.5	30.7	38.1	60.1	53.3	17.4	47.2
GLM-4.6V-Flash	25.9	32.9	42.7	30.9	20.6	19.8
Llama-3.2-11B-V	26.5	34.3	33.3	25.3	21.3	10.4
Gemma-3-12B	30.5	36.9	48.1	41.6	24.4	8.0

PRACTICAL DISCUSSION

Every choice has a cost side

- **Representation:** embedded text plus images matches parser text plus images (50.1% against 49.5%) with fewer tokens. Images cost about 26 s of prefill against 1.8 s.
- **Retriever:** ColQwen3 reaches 81.0% recall at 5 against 58.1% for BM25, but needs a GPU and 110 ms per query against 0.3 ms.
- **Reasoner:** 4-bit keeps accuracy at a smaller footprint (51.8% in about 6.8 GB) but is not faster.

From Findings to Design

The survey and the experiments both point at design principles. I put them together in a [deliberately simple system](#), to see whether they hold up at the level of a whole pipeline rather than one controlled condition.

REPRESENTATION

Keep text and images together

- ▶ Every page read at the full TLV representation, with a strong parser and high-resolution rendering, since the two channels cover for each other's failures.

SELECTION

Gather broadly, present selectively

- ▶ Missing evidence is unrecoverable and moderate distraction is not, so read widely in retrieval-ranked order, but carry most of it forward as compact records rather than as full pages.

REASONING

Check whether the answer is supported

- ▶ Because the reasoner cannot judge sufficiency on its own, a separate step decides whether to accept the answer or keep reading.

No planner, no tool set, no query reformulation. [The point is to validate the principles, not to introduce a complex architecture.](#)

Simple Method Cost

	Turns	Tokens			Latency (s)		Peak
		Input text	Input visual	Output	Prefill	Decode	VRAM (GiB)
reader	11.5	5,030	816	697	1.8	46.8	7.2
synthesizer	1.2	9,307	2,018	463	3.8	34.7	8.1
arbiter	1.2	6,806	1,734	464	2.9	33.3	7.9
Per question, mean	13.8	76,669	13,767	9,092	28.5	617.5	8.2
Per question, median	9.0	43,113	9,830	6,078	15.8	385.0	8.0

- **13.8 turns** per question on average, 11.5 of them the reader.
- The cost is time, not memory: **10.8 minutes** mean, 6.7 median, with peak memory at 8.2 GiB.